RESEARCH POSTER / FOUNDATIONAL MACHINE LEARNING / 2017

ATTENTION IS ALL YOU NEED

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THE CENTRAL IDEA

NO RECURRENCE. ALL ATTENTION.

Model global token interactions directly, while training positions in parallel.

01 / WHY REPLACE RECURRENCE?

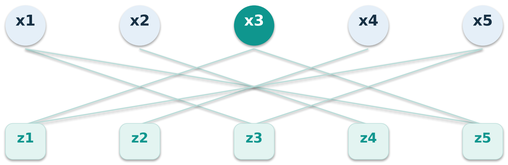
The old bottleneck: each recurrent step waits for the previous one.

RECURRENT / SEQUENTIAL



Sequence depth grows with length n

SELF-ATTENTION / PARALLEL



Each position can attend to other positions in one layer.

02 / THE ATTENTION OPERATION

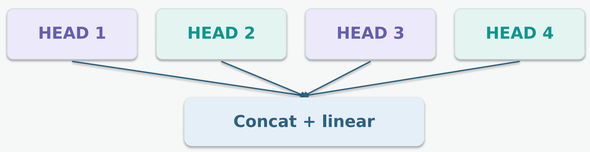
Scaled dot-product attention

$$\text{softmax}(QK^T / \sqrt{d_k})\, V$$

Q queries · K keys · V values

03 / MULTI-HEAD ATTENTION

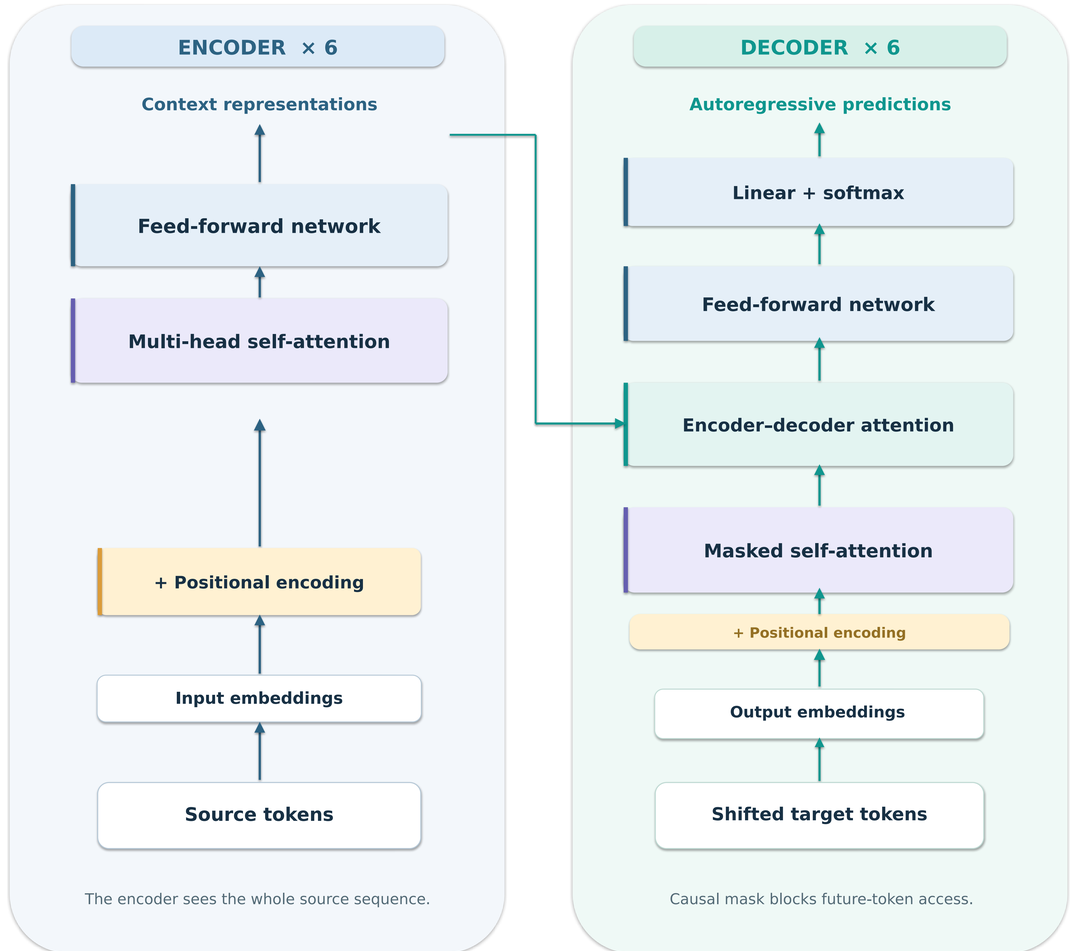
Multiple learned projections capture different relationships.



Schematic: 4 heads shown; base model uses 8.

04 / THE TRANSFORMER, IN ONE VIEW

Two 6-layer stacks. Context flows through attention, not recurrence.



The encoder sees the whole source sequence.

Causal mask blocks future-token access.

Core principle: attention replaces recurrence; residual normalization stabilizes the blocks.

05 / HOW ORDER IS REPRESENTED

Sinusoidal position signals are added to token embeddings.



06 / EVIDENCE FROM TRANSLATION

WMT14 · EN → DE

28.4

BLEU · Transformer (big)

vs 26.36 · ConvS2S ensemble

WMT14 · EN → FR

41.0

BLEU · Transformer (big)

vs 40.56 MoE (single) · 3.5 days / 8 P100 GPUs

07 / BASE-MODEL BLUEPRINT

ENCODER / DECODER	6 / 6 layers	ATTENTION	8 heads
MODEL WIDTH	512 d_model	FEED-FORWARD	2,048 d_ff
PARAMETERS	≈65M base	ATTENTION KEY	64 d_k

08 / WHAT ACTUALLY IMPROVES?

GLOBAL CONTEXT. PARALLEL TRAINING.

Short attention paths between positions.

IMPORTANT LIMITATION

Full self-attention scales quadratically with sequence length. The autoregressive decoder still produces tokens sequentially.

SOURCE / Vaswani et al. (2017), Attention Is All You Need. NeurIPS 2017, Table 2.

Research poster reinterpretation · Architecture redrawn schematically · EN–FR 41.0 uses the 2017 proceedings version (later arXiv v7 reports 41.8).

PAPER